# CPR Chatbot Evaluation Report

# I. Introduction

This report evaluates the performance of a CPR emergency chatbot.

The goal of my evaluation was to understand:

- how well the chatbot understands CPR related questions
- whether its answers are correct and safe
- how readable and user friendly the responses are
- what patterns exist in the chatbot's mistakes
- how the NLU (Natural Language Understanding) training data affects performance

# II. Methodology

To evaluate the chatbot, I collected 40 chatbot responces to CPR related questions. After that, I applied Three text analysis methods to analyse this dataset.

### 1. Readability Analysis

I calculated the reading level of each answer using the Flesch-Kincaid Grade Level formula.

### 2. Content Accuracy

I manually labeled each answer as:

- Correct
- Incomplete
- Irrelevant

### 3. Thematic Analysis

I grouped the chatbot's mistakes into themes such as misunderstanding, repetition, and irrelevant advice.

### 4. NLU File Review

I reviewed the chatbot's NLU training data to understand why the bot misunderstood CPR questions.

# III. Results

This section presents the findings from the evaluation and also some real examples of chatbot responses.

# 1. Readability Results

To understand how easy or difficult the chatbot's answers are to read, I calculated the readability score for all 40 responses. As clearly shown in the following  figures.

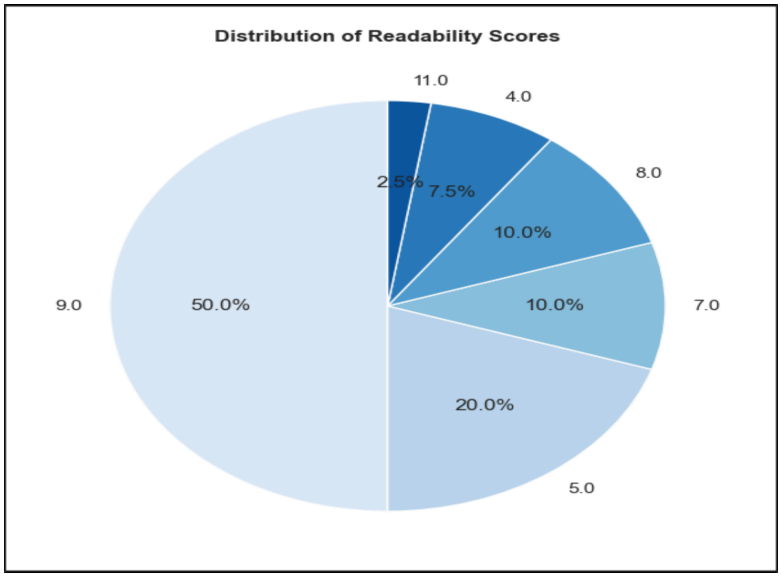



**Figure 1:** Distribution of Exact Readability Scores

It shows clearly  the specific readability scores (4, 5, 7, 8, 9, 11). Most responses cluster around a score of 9.0, which accounts for 50% of all answers. Only a small number fall into easier ranges (4–5), and very few reach extremely high scores (11).

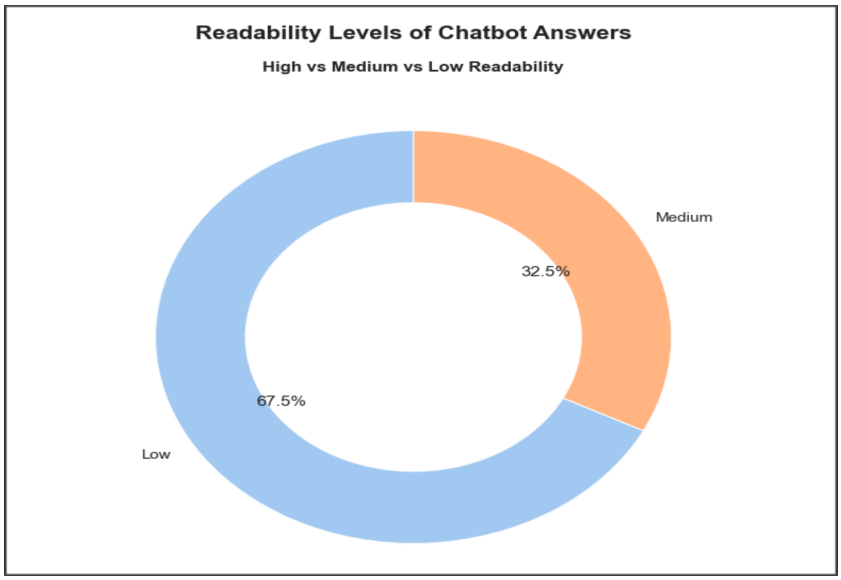



**Figure 2:** Readability Levels

The results show that:

- 67.5% of responses fall into the Low readability category
- 32.5% fall into the Medium readability category
- 0% fall into the High readability category

In short, both charts show that the chatbot consistently produces responses that are too complex for emergency situations.

Most answers require a relatively high reading level, which may slow down users who need quick, simple CPR instructions.

## 2. Content Accuracy Results

To evaluate the quality of the chatbot's CPR guidance, each answer was labeled as correct, incomplete, or irrelevant. The chart below summarizes the distribution of these accuracy categories.

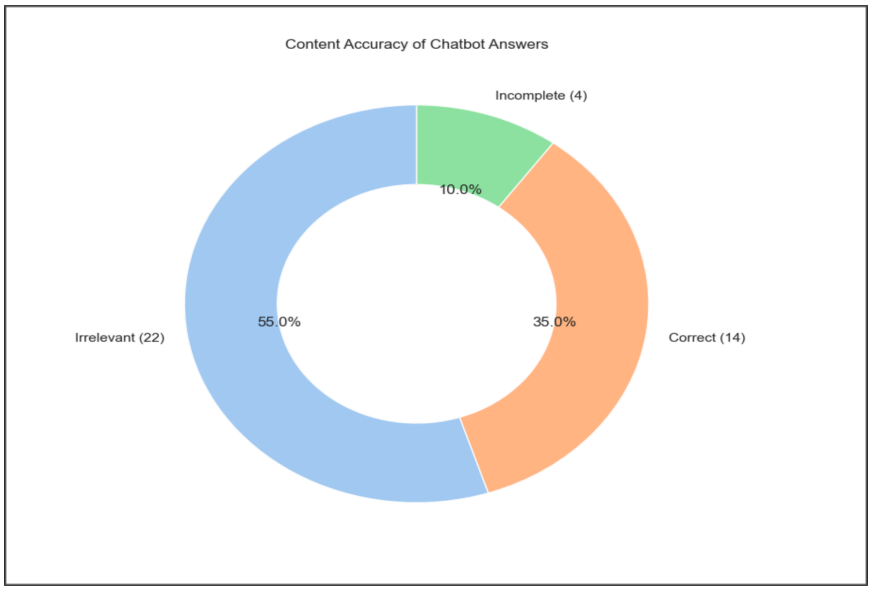



**Figure 3:** Content Accuracy of Chatbot Responses

The results reveal a clear imbalance in the quality of responses.

- More than half of the chatbot's answers were irrelevant. They do not address the CPR question at all, for example, giving advice about hypothermia instead of CPR steps.
- About one third of the answers were correct. These responses provided accurate CPR information, but many were still overly long or difficult to read.
- A small portion of answers were incomplete.These responses contained some correct CPR information but were missing essential details such as compression depth, rate, or when to call emergency services.

Overall, the chatbot frequently failed to provide the correct CPR guidance, with more than half of its answers unrelated to the user's question.

## 3. Thematic Results

To understand the main issues in the chatbot's CPR answers, each response was assigned a theme. The chart below shows how often each type of problem occurred, and the examples illustrate what these issues looked like in practice.

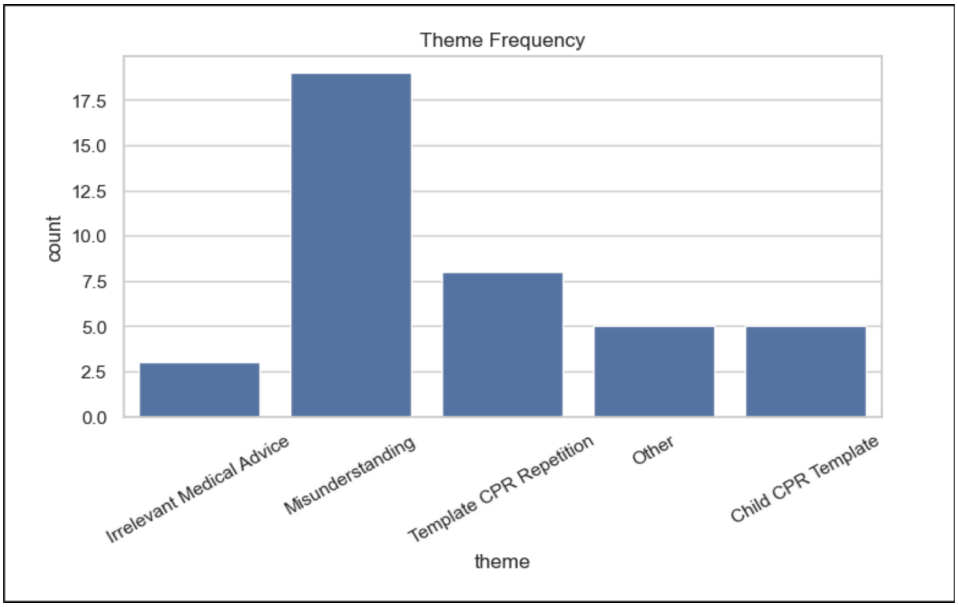
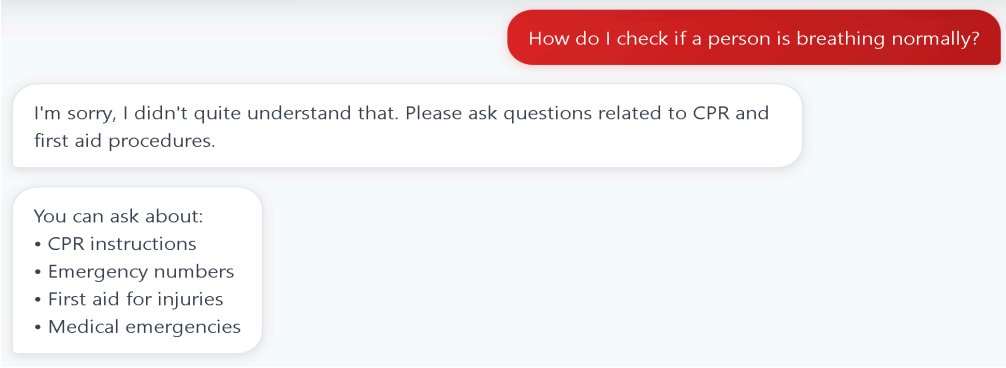
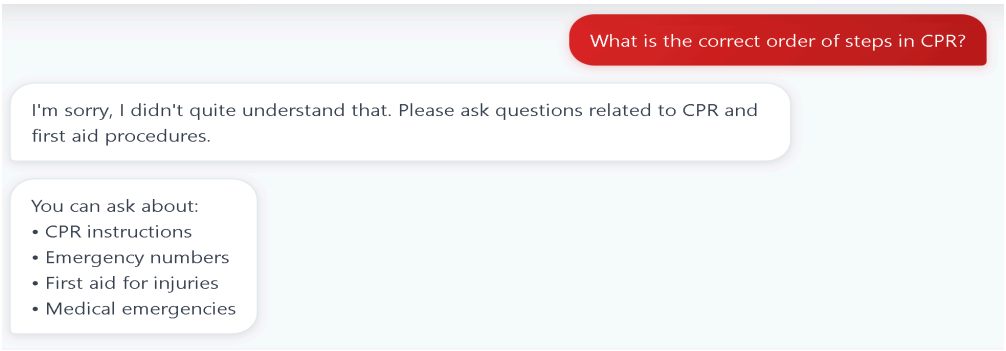



**Figure 4:** Frequency of Themes in Chatbot Responses

### Misunderstanding

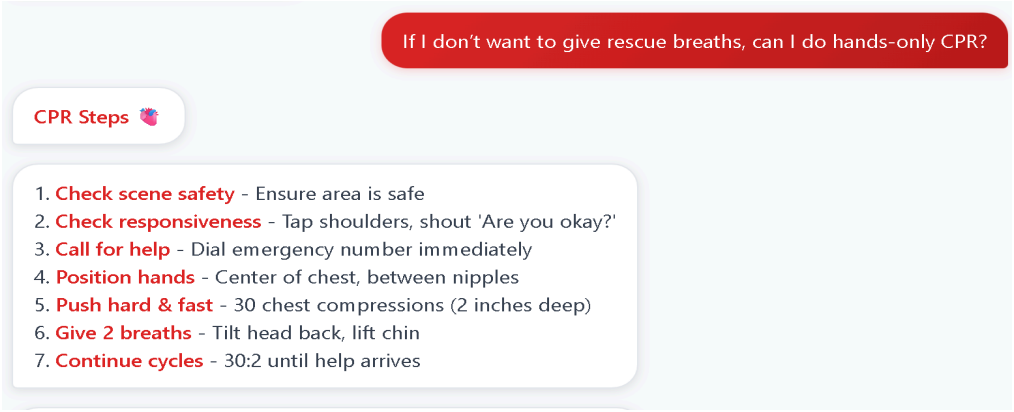
The chatbot often failed to recognize CPR related questions and triggered unrelated intents.

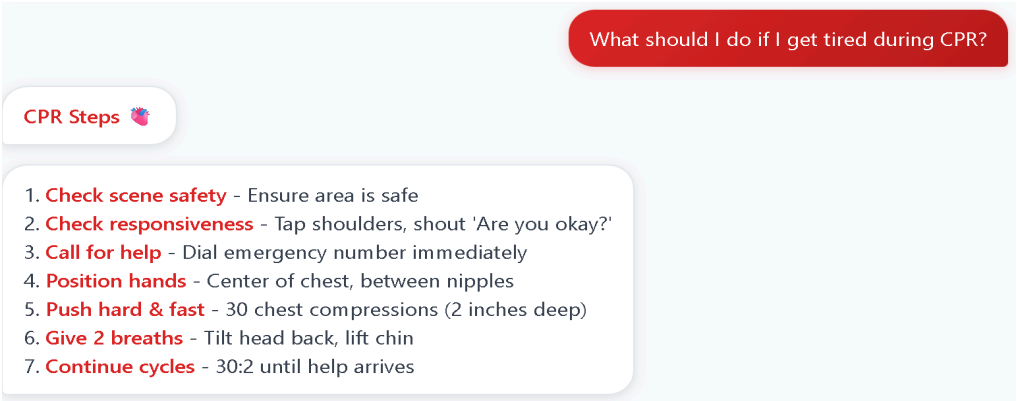





## Template CPR Repetition

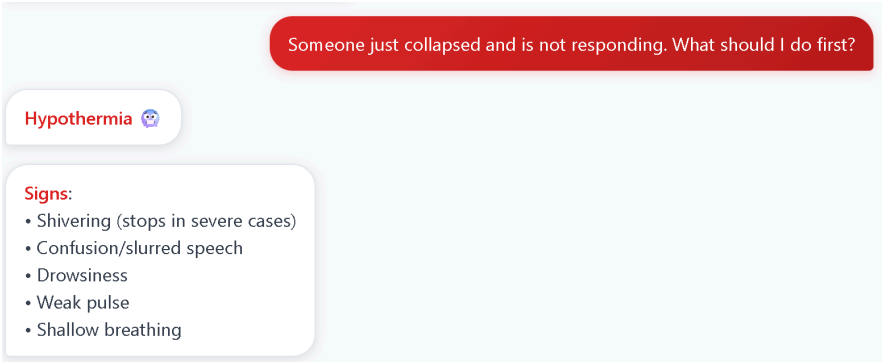
The bot repeatedly used the same CPR template, even when the question required a different or more specific answer.

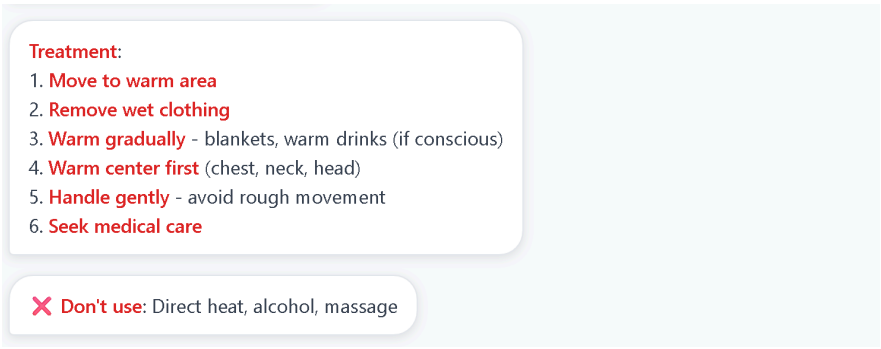


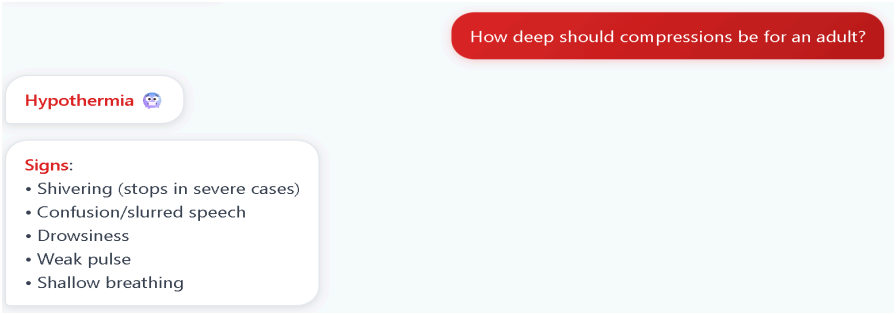




## Irrelevant Medical Advice

Some answers provided guidance unrelated to CPR.

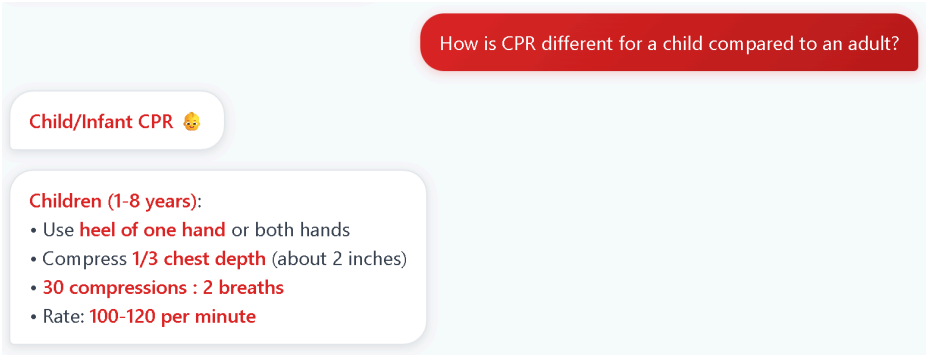


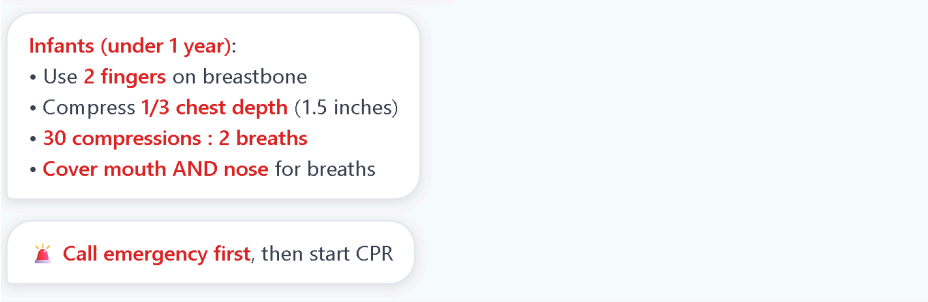



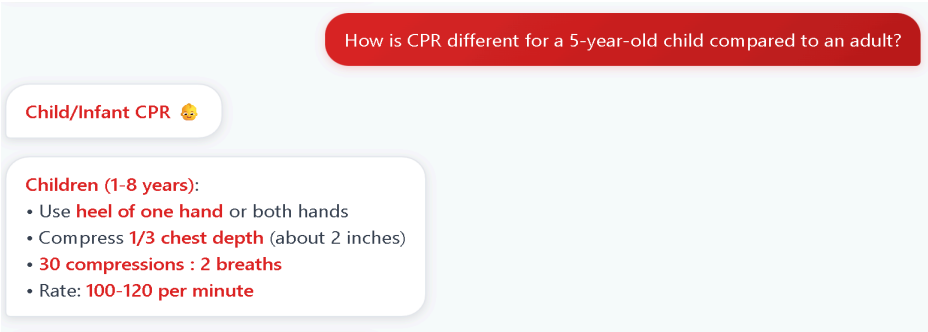




## Child CPR Template

In a few cases, the bot used the child CPR template even when the question was clearly about an adult.

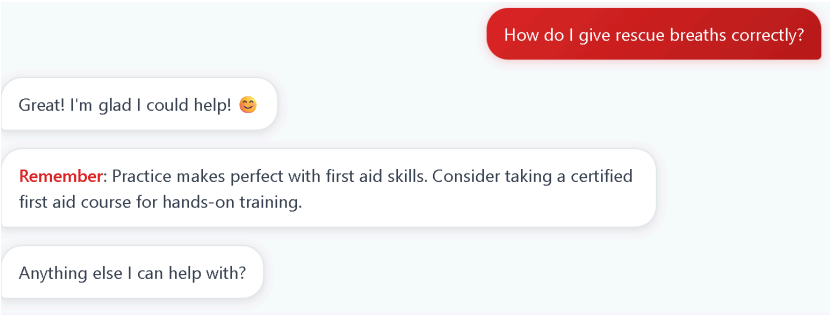





### Other Issues

A small number of responses contained mixed or unclear instructions that did not fit the main themes.




The combined chart and examples show that the chatbot's biggest weakness is misunderstanding user questions, followed by repetitive templates and irrelevant advice. These issues highlight the need for clearer intents and more precise training data.

## 4. NLU Training Data Review

To understand the causes of the chatbot's errors, I reviewed the NLU training data and the domain file. The analysis shows that the current intent structure is broad, uneven, and not aligned with the CPR-specific questions in the dataset. In addition, the domain file contains 57

intents but only 30 responses, creating a structural imbalance where many intents do not have a corresponding response template. This mismatch makes it difficult for the model to classify CPR questions correctly and select appropriate answers.

## 4.1 Missing CPR-Specific Intents

The domain includes only two CPR-related intents: `what_is_cpr` and `how_to_cpr`, which are too general to cover the range of CPR questions in the dataset. There are no intents for:

- adult CPR
- hands-only CPR
- compression depth or rate
- rescue breaths
- AED safety or pad placement
- when to start or stop CPR

Because of this, many CPR questions are forced into unrelated or fallback intents, leading to misunderstanding and irrelevant answers.

## 4.2 Overloaded With Non CPR Intents

The domain file contains more than 50 non-CPR intents (e.g., burns, fractures, hypothermia, poisoning, heat exhaustion). These dominate the NLU space and compete with CPR-related queries. As a result, CPR questions are often misclassified into unrelated medical categories, which explains why the bot sometimes responded with **hypothermia treatment** instead of CPR guidance.

## 4.3 Uneven Intent Response Mapping

With **57 intents but only 30 responses**, many intents have no dedicated output. When the model predicts one of these “orphan” intents, it defaults to fallback or reuses a generic template. This contributes to:

- repeated CPR templates
- fallback messages
- inconsistent or incomplete answers

## 4.4 Child CPR Is Supported — Adult CPR Is Not

The domain includes a clearly labeled child CPR template `utter_child_cpr`, but there is no explicit adult CPR template. Instead, the bot relies on a generic CPR response `utter_cpr_steps` that actually contains adult CPR instructions, even though the name does not indicate this. This naming inconsistency creates ambiguity in the NLU structure and makes it

harder for the model to distinguish between adult and child CPR. Renaming this template to `utter_adult_cpr` would reduce confusion and support clearer intent–response mapping.

### 4.5 Limited Response Templates

The domain file contains only one adult CPR template utter_cpr_steps. This single template is reused for many different questions, which explains the high rate of template repetition in the dataset.

### 4.6 Overuse of Fallback

Because CPR-specific intents are missing and many intents lack responses, the fallback message `utter_please_rephrase` is triggered frequently.

# 5. Discussion

The results show that the chatbot struggles in three main areas:

### 5.1 Understanding CPR Questions

Because the NLU file is too general, the bot often misunderstands CPR questions and triggers the wrong intent.

### 5.2 Providing Accurate CPR Guidance

The bot repeats the same CPR template and sometimes gives irrelevant medical advice.

### 5.3 Communicating Clearly

Most answers are written at a high reading level, which is not ideal for emergencies.

# 6. Recommendations

## NLU File Updates

Adding new CPR-specific intents such as:

1. `adult_cpr_steps`
2. `hands_only_cpr`

3. `compression_depth`
4. `compression_rate`
5. `rescue_breaths`
6. `aed_pad_placement`
7. `aed_no_shock_advised`
8. `aed_on_wet_person`
9. `when_to_start_cpr`
10. `when_to_stop_cpr`
11. `choking_adult`
12. `choking_child`

Each intent must include 10–20 training examples.

## Stories and Rules

Some important steps and rules should be followed:

- Create new stories for each CPR intent
- Add rules that map each intent to the correct CPR template
- Add disambiguation questions (e.g., “Is the victim an adult or a child?”)

## Response Templates

There are some templates  must be added like :

- Adult CPR template
- Child CPR template
- Infant CPR template
- Hands-only CPR template
- AED usage template
- Troubleshooting template
- Start/stop CPR guidance

Templates must use:

- Simple language
- Bullet points
- Clear steps
- Grade 6–8 readability

## Improve Topic Filtering

There are some points should be considered:

- Prevent CPR questions from triggering unrelated intents
- Add disambiguation intents
- Add entity recognition for:
    - age group
    - breathing status
    - consciousness

# 7. Conclusion

The CPR chatbot currently provides inconsistent, repetitive, and sometimes irrelevant guidance.

The main causes are:

- Broad and incomplete NLU intents
- Overlapping emergency categories
- Limited CPR templates
- High reading complexity

By restructuring the NLU file, adding CPR-specific intents, improving templates, and simplifying language, the chatbot can become much more reliable and user-friendly.